



### New neuromorphic chip design

Researchers have come up with new chip designs for processors working similar to our brains. <http://dgit.in/MITbmchip>



### Chrome 64 update on Android

Chrome's latest update which removes harmful pop-ups and redirects is hitting Android first. <http://dgit.in/Chrm64ads>



a password manager but it also supports facial recognition. It works like regular password managers by installing an extension on your browser and making logins secure and super fast. Your facial details are stored on their cloud servers and you can add more than one factor to authenticate access. So, if you're logging into True Key on a new browser, you can use your facial details, a registered device and a master password, all together to authenticate a single secure login. True Key is supported on mobile devices as well.

**True Key:** <http://dgit.in/IntlTrueKy>

### FACE AND HEAD TRACKING

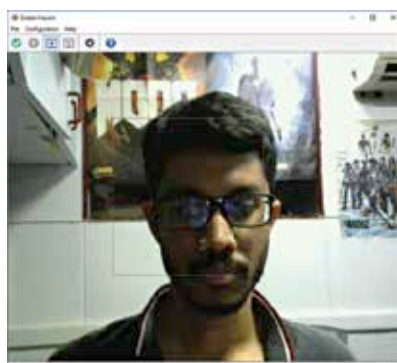
Virtual reality is taking over gaming rapidly, however, owning a VR headset is still an expensive ordeal. What if we told you that you could build a complete VR headset for under ₹1,000? A few years ago, when VR had just entered the market, we set out to find out whether we could build a cheap VR headset that was capable of offering all the functionalities of a mainstream one. We were successful in creating our own "Oculus Thrift". The entire contraption was made with an accelerometer and LEDs to track our head movement in tandem with the webcam. A face tracking software called Face Track 2.0 was used along with the hardware. This setup allowed us to play a few video games back in 2014. And now, more games have been added. If you're already interested, you should surely check out the entire process here: <http://dgit.in/OculusThrift>.

**FaceTrackNoIR:** <http://dgit.in/FTNoIR>



### DESKTOP CONTROL

It's easier to control the desktop with your mouse and keyboard, but it's definitely cool to control it with your head. Using Viacam, you can move your head to control the mouse. The software dynamically tracks your face by projecting data points on your face and accordingly moves the mouse based on your head movement. If you want to select something, you allow the mouse to stay over the button or icon for a second. It certainly isn't a practical solution, no matter how much you calibrate it. Viacam is free to use and requires a



good amount of light falling on your face. You can change the acceleration and speed according to your preferences so that you don't have to move your head too much.

**Viacam:** <http://dgit.in/viacamhead>

### GESTURE CONTROL FOR MUSIC PLAYBACK

Do you still use your mouse or keyboard to pause or skip music? All of the above can be done using hand gestures. An app called Flutter recognises your hand and a set of gestures which can be used to play, pause, and play the previous or next song. There aren't any settings in the app, all you need to do is turn the camera on through the app to recognise your hand gestures. Placing your palm in front of the camera either plays or pauses the song. In order to play the previous or next song, you need to do a horizontal thumbs-up gesture. Based on the direction your thumb is pointing,



the song changes. So, pointing your thumb to the left, causes the previous song to start playing while pointing your thumb to the right skips the current song. In order to recognise your hand gestures, you need to keep your hand at a recommended distance of 1-6 feet. There haven't been any announcements or updates from the developers since 2013 after they were acquired by Google.

**Flutter:** <http://dgit.in/FlutterApp>

### GESTURE CONTROL FOR GAMES

Now, don't expect to be able to control just about any video game out there with your hands. Camspaced recognises objects and lets you use them as controllers in supported games. Although the software claims to support any external game with a virtual controller, we couldn't get it up and running. However, we were able to play the games within the software such as Air Hockey and Curve Ball. You need a coloured ob-



ject (bright colour recommended) to be recognised as a controller which has to be moved around to control the game. You also need to ensure that you have suitable lighting over the object. Tracking isn't perfect but it is definitely fun to play some games using gestures.

**Camspaced:** <http://dgit.in/camspacedev>